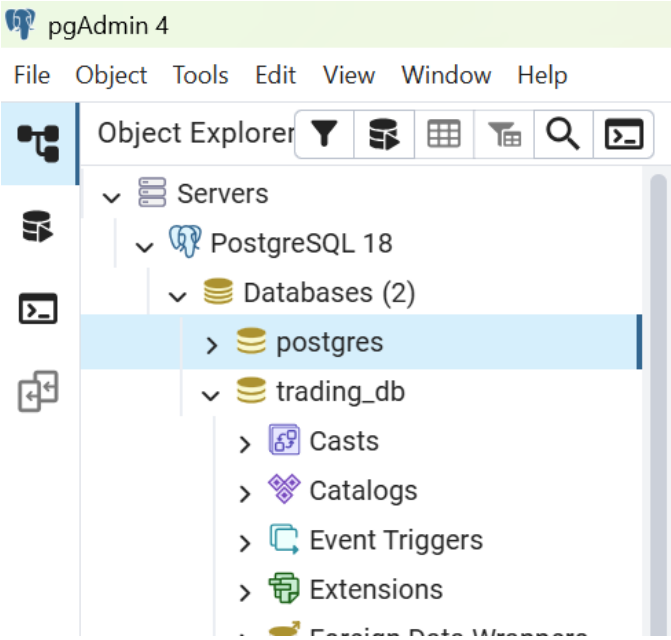
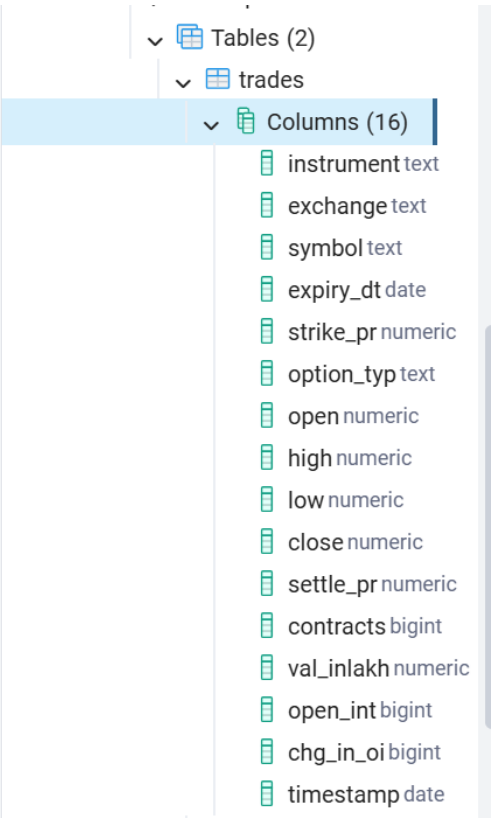
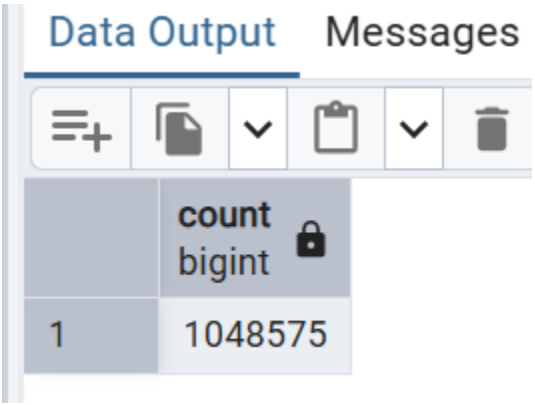
1. Database Setup



2. Table Creation



3. Data Import



Data Output Messages

	count bigint
1	1048575

4. SQL Analysis Queries

Query 1 – Top Symbols by OI Change

```sql
--Top 10 symbols by open interest (OI) change across exchanges.
--This shows which trading symbols had the biggest change in Open Interest.
-- Rank symbols by trading volume
SELECT
    symbol,
    SUM(contracts) AS total_volume,
    RANK() OVER (ORDER BY SUM(contracts) DESC) AS volume_rank
FROM trades
GROUP BY symbol
ORDER BY volume_rank
LIMIT 10;
```

Output:

	symbol text	total_volume numeric	volume_rank bigint
1	BANKNIFTY	432403426	1
2	NIFTY	182874865	2
3	YESBANK	3862506	3
4	RELIANCE	2679721	4
5	IBULHSGF...	2060755	5
6	SBIN	1990085	6
7	MARUTI	1667629	7
8	ICICIBANK	1581757	8
9	HDFCBANK	1462647	9
10	TATASTEEL	1347950	10

Query 2 – Volatility Analysis

```sql
--Volatility analysis: 7-day rolling std dev of close prices for NIFTY options.
--This calculates price volatility for NIFTY options.
-- 7-day rolling volatility for NIFTY options
SELECT
    symbol,
    timestamp,
    close,
    STDDEV(close) OVER (
        PARTITION BY symbol
        ORDER BY timestamp
        ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
    ) AS volatility_7_day
FROM trades
WHERE symbol = 'NIFTY'
AND instrument = 'OPTIDX'
ORDER BY timestamp;
```

Output:

	symbol text	timestamp date	close numeric	volatility_7_day numeric
1	NIFTY	2019-08-01	2496.5	[null]
2	NIFTY	2019-08-01	118.2	1681.712057695966
3	NIFTY	2019-08-01	126.25	1370.794219604095
4	NIFTY	2019-08-01	134.7	1185.077481556234
5	NIFTY	2019-08-01	143.5	1058.076739655494
6	NIFTY	2019-08-01	152.75	964.123075130971
7	NIFTY	2019-08-01	162.4	890.939846387483
8	NIFTY	2019-08-01	172.45	19.5415978778360809
9	NIFTY	2019-08-01	182.9	20.4091832738295019
10	NIFTY	2019-08-01	193.85	21.3043590442358600

Due to the large dataset, only the first 10 rows of the volatility analysis output are shown for demonstration.

Query 3 – Cross Exchange Comparisons

```sql
--Cross-exchange comparison: Avg settle_pr for gold futures (MCX) vs. equity index futures (NSE).
--This compares average closing prices across exchanges like NSE, BSE, MCX.
SELECT
    exchange,
    AVG(close) AS avg_close_price
FROM trades
GROUP BY exchange;
```

Output:

	exchange text	avg_close_price numeric
1	NSE	311.3572659561786234

Query 4 : Option Chain Summary

```
--Option chain summary: Grouped by expiry_dt and strike_pr, calculating implied volume.
--This summarizes trading activity by expiry date.
SELECT
    symbol,
    expiry_dt,
    SUM(contracts) AS total_contracts
FROM trades
GROUP BY symbol, expiry_dt
ORDER BY expiry_dt;
```

Output

	symbol text	expiry_dt date	total_contracts numeric
1	BANKNIFTY	2019-08-01	30146563
2	NIFTY	2019-08-01	8061286
3	NIFTYIT	2019-08-01	1
4	BANKNIFTY	2019-08-08	67963829
5	NIFTY	2019-08-08	28862687
6	NIFTYIT	2019-08-08	0
7	BANKNIFTY	2019-08-14	18954743
8	NIFTY	2019-08-14	11182515
9	NIFTYIT	2019-08-14	0
10	BANKNIFTY	2019-08-22	269613

Due to the large dataset, only the first 10 rows of the option chain summary output are shown for demonstration.

Query 5 : Performance Optimized Query

```
--Performance-optimized query for max volume in last 30 days using indexes/window functions.
--Finds highest traded symbols.
SELECT
    symbol,
    MAX(contracts) AS max_volume
FROM trades
GROUP BY symbol
ORDER BY max_volume DESC
LIMIT 10;
```

Output

	symbol text	max_volume bigint
1	BANKNIFTY	3474323
2	NIFTY	1359253
3	YESBANK	175291
4	RELIANCE	132503
5	IBULHSGFIN	78587
6	SBIN	65130
7	INDUSIND...	59763
8	ICICIBANK	54549
9	TATASTEEL	53188
10	HDFC	53088

Query Optimization

```
--Query Optimization
EXPLAIN ANALYZE
SELECT MAX(contracts)
FROM trades;
```

Output:

	QUERY PLAN text
1	Finalize Aggregate (cost=23141.54..23141.55 rows=1 width=8) (actual time=153.999..163.774 rows=1.00 loops=1)
2	Buffers: shared hit=7895 read=8785
3	-> Gather (cost=23141.33..23141.54 rows=2 width=8) (actual time=153.796..163.766 rows=3.00 loops=1)
4	Workers Planned: 2
5	Workers Launched: 2
6	Buffers: shared hit=7895 read=8785
7	-> Partial Aggregate (cost=22141.33..22141.34 rows=1 width=8) (actual time=105.130..105.131 rows=1.00 loops=3)
8	Buffers: shared hit=7895 read=8785
9	-> Parallel Seq Scan on trades (cost=0.00..21049.06 rows=436906 width=8) (actual time=1.242..47.866 rows=349525.00 loo...
10	Buffers: shared hit=7895 read=8785
11	Planning Time: 0.099 ms
12	Execution Time: 163.810 ms